

Лабораторная работа № 2

Настройка DNS-сервера

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1. Информация

1.1 Докладчик

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2. Цель и задание

2.1 Цель работы

Получить практические навыки установки и настройки DNS-сервера BIND, создания прямой и обратной зон, проверки разрешения доменных имён и автоматизации конфигурации серверной виртуальной машины средствами Vagrant.

2.2 Основные задачи

- установить и настроить BIND;
- создать прямую и обратную зоны;
- настроить NetworkManager, firewalld и SELinux;
- проверить DNS с помощью dig и host;
- автоматизировать настройку в Vagrant.

3. Выполнение лабораторной работы

3.1 Подготовка виртуальной машины

Запущена виртуальная машина
server.

```
Far Manager, version 3.0.6699.0 x86
Copyright © 1996-2000 Eugene Roshal, Copyright © 2000-2026 Far Group
C:\work\rapavlichenko>vagrant up server
Bringing machine 'server' up with 'virtualbox' provider...
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: Clearing any previously set forwarded ports...
==> server: Clearing any previously set network interfaces...
==> server: Preparing network interfaces based on configuration...
server: Adapter 1: nat
server: Adapter 2: intnet
==> server: Forwarding ports...
```

Выполнено подключение по SSH.

```
C:\work\rapavlichenko>vagrant ssh server
==> server: The machine you're attempting to SSH into is configured to use
==> server: password-based authentication. Vagrant can't script entering the
==> server: password for you. If you're prompted for a password, please ente
r
==> server: the same password you have configured in the Vagrantfile.
vagrant@127.0.0.1's password: |
```

3.2 Переход к административной работе

Выполнен вход под
пользователем rapavlichenko.

```
vagrant@server:~$ su - rapavlichenko
Password:
Last login: Fri Sep 11 19:09:13 UTC 2026 on pts/0
[rapavlichenko@server.rapavlichenko.net ~]$
```

Получены полномочия
суперпользователя.

```
[rapavlichenko@server.rapavlichenko.net ~]$ sudo -i
```

We trust you have received the usual lecture from the local System Administrator. It usually boils down to these three things:

- #1) Respect the privacy of others.
- #2) Think before you type.
- #3) With great power comes great responsibility.

For security reasons, the password you type will not be visible.

```
[sudo] password for rapavlichenko:
```

```
[root@server.rapavlichenko.net ~]#
```

3.3 Установка BIND

Установлены пакеты bind и bind-utils, содержащие DNS-сервер и диагностические утилиты.

```
[root@server.rapavlichenko.net ~]# dnf -y install bind bind-utils
Extra Packages for Enterprise Linux 10 - x86_64 6.0 kB/s | 39 kB    00:06
Extra Packages for E    [          ===          ] ---  B/s |  0 B    --:-- ETA
```

3.4 Проверка исходной работы DNS

Команда `dig` вернула А-записи внешнего узла. Одновременно выявлен недоступный DNS-сервер 172.19.0.2.

```
[root@server.rapavlichenko.net ~]# dig www.yandex.ru
;; communications error to 172.19.0.2#53: timed out
;; communications error to 172.19.0.2#53: timed out
;; communications error to 172.19.0.2#53: timed out

; <<>> DiG 9.18.33 <<>> www.yandex.ru
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 57499
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4000
;; QUESTION SECTION:
;www.yandex.ru.                IN      A

;; ANSWER SECTION:
www.yandex.ru.                83      IN      A      5.255.255.77
www.yandex.ru.                83      IN      A      77.88.55.88
www.yandex.ru.                83      IN      A      77.88.44.55
```

3.5 Основные файлы конфигурации

/etc/resolv.conf задаёт используемые DNS-серверы.

```
[rapavlichenko@server.rapavlichenko.net ~]$ cat /etc/resolv.conf
# Generated by NetworkManager
search rapavlichenko.net
nameserver 172.19.0.2
nameserver 192.168.150.36
nameserver 192.168.150.37
# NOTE: the libc resolver may not support more than 3 nameservers.
# The nameservers listed below may not be recognized.
nameserver fd17:625c:f037:2::3
[rapavlichenko@server.rapavlichenko.net ~]$
```

/etc/named.conf задаёт параметры BIND.

```
//
// named.conf
//
// Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
// server as a caching only nameserver (as a localhost DNS resolver only).
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//

options {
    listen-on port 53 { 127.0.0.1; };
    listen-on-v6 port 53 { ::1; };
    directory "/var/named";
    dump-file "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named_stats.txt";
    memstatistics-file "/var/named/data/named_mem_stats.txt";
    secroots-file "/var/named/data/named.secroots";
    recursing-file "/var/named/data/named.recursing";
    allow-query { localhost; };

    /*
     * If you are building a AUTHORITATIVE DNS server, do NOT enable recursion.
     * If you are building a RECURSIVE (caching) DNS server, you need to enable
     * recursion.
     * If your recursive DNS server has a public IP address, you MUST enable access
     * control to limit queries to your legitimate users. Failing to do so will
     * cause your server to become part of large scale DNS amplification
     * attacks. Implementing BCP38 within your network would greatly
     * reduce such attack surface
     */
    recursion yes;

    dnssec-validation yes;

    managed-keys-directory "/var/named/dynamic";
    geoip-directory "/usr/share/GeoIP";

    pid-file "/run/named/named.pid";
    session-keyfile "/run/named/session.key";

    /* https://fedoraproject.org/wiki/Changes/CryptoPolicy */
    include "/etc/crypto-policies/back-ends/bind.config";
};

logging {
    channel default_debug {
        file "data/named.run";
        severity dynamic;
    };
};

zone "." IN {
    type hint;
    file "named.ca";
};
```

3.6 Шаблоны файлов зон

Шаблон прямой локальной зоны.

```
[root@server.rapavlichenko.net ~]# cat /var/named/named.localhost
$TTL 1D
@      IN SOA  @ rname.invalid. (
                                0      ; serial
                                1D     ; refresh
                                1H     ; retry
                                1W     ; expire
                                3H )   ; minimum

NS     @
A      127.0.0.1
AAAA   ::1
```

Шаблон обратной локальной зоны.

```
[root@server.rapavlichenko.net ~]# cat /var/named/named.loopback
$TTL 1D
@      IN SOA  @ rname.invalid. (
                                0      ; serial
                                1D     ; refresh
                                1H     ; retry
                                1W     ; expire
                                3H )   ; minimum

NS     @
A      127.0.0.1
AAAA   ::1
PTR     localhost.
[root@server.rapavlichenko.net ~]#
```

3.7 Корневые DNS-серверы

Файл `named.ca` содержит адреса корневых DNS-серверов, с которых начинается рекурсивное разрешение имён.

```
[root@server.rapavlichenko.net ~]# cat /var/named/named.ca
;
; This file holds the information on root name servers needed to
; initialize cache of Internet domain name servers
; (e.g. reference this file in the "cache . <file>"
; configuration file of BIND domain name servers).
;
; This file is made available by InterNIC
; under anonymous FTP as
;   file      /domain/named.cache
;   on server  FTP.INTERNIC.NET
; -OR-       RS.INTERNIC.NET
;
; last update:   December 20, 2023
; related version of root zone:  2023122001
;
; FORMERLY NS.INTERNIC.NET
;
;
;      3600000      NS      A.ROOT-SERVERS.NET.
A.ROOT-SERVERS.NET. 3600000      A      198.41.0.4
A.ROOT-SERVERS.NET. 3600000      AAAA   2001:503:ba3e::2:30
;
; FORMERLY NS1.ISI.EDU
;
;      3600000      NS      B.ROOT-SERVERS.NET.
B.ROOT-SERVERS.NET. 3600000      A      170.247.170.2
B.ROOT-SERVERS.NET. 3600000      AAAA   2801:1b8:10::b
;
; FORMERLY C.PSI.NET
;
;      3600000      NS      C.ROOT-SERVERS.NET.
C.ROOT-SERVERS.NET. 3600000      A      192.33.4.12
C.ROOT-SERVERS.NET. 3600000      AAAA   2001:500:2::c
;
; FORMERLY TERP.UMD.EDU
;
;      3600000      NS      D.ROOT-SERVERS.NET.
D.ROOT-SERVERS.NET. 3600000      A      199.7.91.13
D.ROOT-SERVERS.NET. 3600000      AAAA   2001:500:2d::d
;
; FORMERLY NS.NASA.GOV
;
;      3600000      NS      E.ROOT-SERVERS.NET.
E.ROOT-SERVERS.NET. 3600000      A      192.203.230.10
E.ROOT-SERVERS.NET. 3600000      AAAA   2001:600:28::e
```

3.8 Первый запуск и диагностика

Служба named была запущена.

```
[root@server.rapavlichenko.net ~]# systemctl start named  
[root@server.rapavlichenko.net ~]#
```

Проверка показала, что локальный DNS ещё не используется системой.

```
[root@server.rapavlichenko.net ~]# dig www.yandex.ru  
;; communications error to 172.19.0.2#53: timed out  
;; communications error to 172.19.0.2#53: timed out  
;; communications error to 172.19.0.2#53: timed out  
  
;<>> DiG 9.18.33 <>> www.yandex.ru  
;; global options: +cmd  
;; Got answer:  
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 9373  
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
;; EDNS: version: 0, flags:; udp: 4000  
;; QUESTION SECTION:  
;www.yandex.ru.                IN      A  
  
;; ANSWER SECTION:  
www.yandex.ru.                161     IN      A      77.88.44.55  
www.yandex.ru.                161     IN      A      5.255.255.77  
www.yandex.ru.                161     IN      A      77.88.55.88  
  
;; Query time: 13 msec  
;; SERVER: 192.168.150.36#53(192.168.150.36) (UDP)  
;; WHEN: Tue Sep 15 20:04:43 UTC 2026  
;; MSG SIZE rcvd: 90  
  
[root@server.rapavlichenko.net ~]# dig @127.0.0.1 www.yandex.ru  
  
;<>> DiG 9.18.33 <>> @127.0.0.1 www.yandex.ru  
;; (1 server found)  
;; global options: +cmd  
;; Got answer:  
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 12356  
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
;; EDNS: version: 0, flags:; udp: 1232  
;; COOKIE: cbbai9d7ee6d177c010000006aa9a4f41ff7c61d04049307 (good)  
;; QUESTION SECTION:  
;www.yandex.ru.                IN      A  
  
;; Query time: 4071 msec  
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
```

3.9 Открытие профиля eth0

Запущен интерактивный редактор NetworkManager.

```
[root@server.rapavlichenko.net ~]# nmcli connection edit eth0
```

Выбран профиль подключения eth0.

```
[root@server.rapavlichenko.net ~]# nmcli connection edit eth0
```

3.10 Изменение DNS в NetworkManager

Удалены автоматически полученные DNS-серверы, включён ignore-auto-dns и задан адрес 127.0.0.1.

```
nmcli> remove ipv4.dns
nmcli> set ipv4.ignore-auto-dns yes
nmcli> set ipv4.dns 127.0.0.1
nmcli> save
Connection 'eth0' (31ef6182-2c55-44b3-8944-a2af7d10d59a) successfully updated.
nmcli> quit
[root@server.rapavlichenko.net ~]#
```

3.11 Настройка локального DNS

Для профиля eth0 отключено получение DNS по DHCP и задан локальный сервер 127.0.0.1.

```
[root@server.rapavlichenko.net ~]# nmcli connection edit eth0

===| nmcli interactive connection editor |===

Editing existing '802-3-ethernet' connection: 'eth0'

Type 'help' or '?' for available commands.
Type 'print' to show all the connection properties.
Type 'describe [<setting>.<prop>]' for detailed property description.

You may edit the following settings: connection, 802-3-ethernet (ethernet), 802-1x, dcb,
, tc, proxy
nmcli> remove ipv4.dns
nmcli> set ipv4.ignore-auto-dns yes
nmcli> set ipv4.dns 127.0.0.1
nmcli> save
Connection 'eth0' (31ef6182-2c55-44b3-8944-a2af7d10d59a) successfully updated.
nmcli> quit
[root@server.rapavlichenko.net ~]#
```

3.12 Проверка resolv.conf

После перезапуска NetworkManager локальный DNS-сервер появился в /etc/resolv.conf.

```
[root@server.rapavlichenko.net ~]# systemctl restart NetworkManager
[root@server.rapavlichenko.net ~]# cat /etc/resolv.conf
cat: /etc/resolv.conf: No such file or directory
[root@server.rapavlichenko.net ~]# cat /etc/resolv.conf
# Generated by NetworkManager
search rapavlichenko.net
nameserver 127.0.0.1
nameserver fd17:625c:f037:2::3
[root@server.rapavlichenko.net ~]#
```

3.13 Настройка BIND и firewalld

Разрешены запросы из локальной сети.

```
GNU nano 8.1 /etc/named.conf
//
// named.conf
//
// Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
// server as a caching only nameserver (as a localhost DNS resolver only).
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//

options {
    listen-on port 53 { 127.0.0.1; any; };
    listen-on-v6 port 53 { ::1; };
    directory "/var/named";
    dump-file "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named_stats.txt";
    memstatistics-file "/var/named/data/named_mem_stats.txt";
    secroots-file "/var/named/data/named.secrets";
    recursing-file "/var/named/data/named.recursing";
    allow-query { localhost; 192.168.0.0/16; };

/*
- If you are building an AUTHORITATIVE DNS server, do NOT enable recursion.
- If you are building a RECURSIVE (caching) DNS server, you need to enable
  recursion.
- If your recursive DNS server has a public IP address, you MUST enable access
  control to limit queries to your legitimate users. Failing to do so will
  cause your server to become part of large scale DNS amplification
  attacks. Implementing BCP38 within your network would greatly
  reduce such attack surface
*/
recursion yes;
```

В firewalld разрешена служба DNS.

```
[root@server.rapavlichenko.net ~]# firewall-cmd --add-service=dns
success
[root@server.rapavlichenko.net ~]# firewall-cmd --add-service=dns --permanent
success
[root@server.rapavlichenko.net ~]#
```

3.14 Проверка сетевого порта

Командой `lsof` проверено использование UDP-портов. Служба DNS использует порт 53.

```
[root@server.rapavlichenko.net ~]# lsof | grep UDP
lsof: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1001/gvfs
Output information may be incomplete.
lsof: WARNING: can't stat() fuse.portal file system /run/user/1001/doc
Output information may be incomplete.
systemd      1                root    339u      IPv4            6151      0t0      UDP *:sunrpc
systemd      1                root    341u      IPv6            6167      0t0      UDP *:sunrpc
avahi-daemon 1065            avahi    12u      IPv4            9870      0t0      UDP *:mdns
avahi-daemon 1065            avahi    13u      IPv6            9871      0t0      UDP *:mdns
chronyd      1066            chrony    4u      IPv4           10109      0t0      UDP localhost:323
chronyd      1066            chrony    5u      IPv6           10110      0t0      UDP localhost:323
chronyd      1066            chrony    6u      IPv4           10111      0t0      UDP *:ntp
kea-dhcpd    1527            kea      10u      IPv4           13012      0t0      UDP localhost:53001
kea-dhcpd    1527 1715 kea-dhcpd    kea      10u      IPv4           13012      0t0      UDP localhost:53001
```

3.15 Подготовка пользовательского файла зон

Скопирован файл примеров зон.

```
[root@server.rapavlichenko.net ~]# cp /etc/named.rfc1912.zones /etc/named
[root@server.rapavlichenko.net ~]# cp /etc/named
cp: missing destination file operand after '/etc/named'
Try 'cp --help' for more information.
[root@server.rapavlichenko.net ~]# cp /etc/named
cp: missing destination file operand after '/etc/named'
Try 'cp --help' for more information.
[root@server.rapavlichenko.net ~]# cd /etc/named
[root@server.rapavlichenko.net named]# mv /etc/named/named.rfc1912.zones /etc/named/rapavlichenko.net
[root@server.rapavlichenko.net named]#
```

Файл зон подключён в named.conf.

```
zone "." IN {
    type hint;
    file "named.ca";
};

include "/etc/named.rfc1912.zones";
include "/etc/named.root.key";

include "/etc/named/user.net"
```

3.16 Описание зон

В конфигурацию подключены прямая зона `garavlichenko.net` и обратная зона `1.168.192.in-addr.arpa`.

[illegible]

3.17 Каталоги файлов зон

Созданы каталоги master/fz для прямых зон и master/rz для обратных зон.

```
[root@server.rapavlichenko.net named]# cd /var/named  
[root@server.rapavlichenko.net named]# mkdir -p /var/named/master/fz  
[root@server.rapavlichenko.net named]# mkdir -p /var/named/master/rz  
[root@server.rapavlichenko.net named]#
```

3.18 Подготовка прямой зоны

Шаблон `named.localhost` скопирован в каталог прямых зон и переименован в `rapavlichenko.net`.

```
[root@server.rapavlichenko.net named]# cp /var/named/named.localhost /var/named/master/fz
[root@server.rapavlichenko.net named]# cd /var/named/master/fz/
[root@server.rapavlichenko.net fz]# mv named.localhost user.net
[root@server.rapavlichenko.net fz]# mv named.localhost rapavlichenko.net
mv: cannot stat 'named.localhost': No such file or directory
[root@server.rapavlichenko.net fz]# mv user.net rapavlichenko.net
[root@server.rapavlichenko.net fz]# █
```

3.19 Прямая зона

Созданы SOA- и NS-записи, а также A-записи узлов server и ns с адресом 192.168.1.1.

```
GNU nano 8.1 /var/named/master/fz/rapavlichenko.net
$TTL 1D
@      IN SOA  @ server.rapavlichenko.net. (
                                2026091500 ; serial
                                1D         ; refresh
                                1H         ; retry
                                1W         ; expire
                                3H )       ; minimum
      NS   @
      A    192.168.1.1
$ORIGIN rapavlichenko.net.
server A    192.168.1.1
ns     A    192.168.1.1
```

3.20 Подготовка обратной зоны

Шаблон `named.loopback` скопирован в каталог обратных зон и переименован в `192.168.1`.

```
root@server:~# mkdir -p /var/named/master/rz
root@server:~# cp /var/named/named.loopback /var/named/master/rz/
root@server:~# cd /var/named/master/rz
root@server:/var/named/master/rz# mv named.loopback 192.168.1
root@server:/var/named/master/rz# nano 192.168.1
```

3.21 Обратная зона

PTR-записи связывают адрес 192.168.1.1 с именами server.rapavlichenko.net и ns.rapavlichenko.net.

```
GNU nano 8.1
$TTL 1D
@      IN SOA  @ server.rapavlichenko.net. (
                                2026091500 ; serial
                                1D          ; refresh
                                1H          ; retry
                                1W          ; expire
                                3H )        ; minimum
      NS   @
      A    192.168.1.1
      PTR  server.rapavlichenko.net.
$ORIGIN 1.168.192.in-addr.arpa.
1      PTR  server.rapavlichenko.net.
1      PTR  ns.rapavlichenko.net.|
```

3.22 Настройка SELinux

Восстановлены контексты файлов и включён параметр
`named_write_master_zones`.

```
root@server:~# chcon -R named_t named /etc/named
root@server:~# chcon -R named_t named /var/named
root@server:~# restorecon -v /etc
Relabeled /etc/NetworkManager/system-connections/eth1.mconnection from unconfined_u:object_r:user_tcp_t:s0 to unconfined_u:object_r:NetworkManager_etc_rw_t:s0
root@server:~# restorecon -v /var/named
root@server:~#
```

```
root@server:~# getsebool -a | grep named
named_tcp_bind_http_port --> off
named_write_master_zones --> on
root@server:~# setsebool named_write_master_zones 1
setsebool -P named_write_master_zones 1
```

3.23 Проверка системного журнала

В журнале systemd проверена работа служб после изменения конфигурации.

```
vagrant@server:~$ sudo -i
root@server:~# journalctl -x -f
Sep 15 21:06:24 server.rapavlichenko.net systemd[1]: serial-getty@ttyS0.service: De
Subject: Unit succeeded
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

The unit serial-getty@ttyS0.service has successfully entered the 'dead' state.
Sep 15 21:06:25 server.rapavlichenko.net systemd[1]: serial-getty@ttyS0.service: Sc
Subject: Automatic restarting of a unit has been scheduled
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

Automatic restarting of the unit serial-getty@ttyS0.service has been scheduled,
the configured Restart= setting for the unit.
Sep 15 21:06:25 server.rapavlichenko.net systemd[1]: Started serial-getty@ttyS0.ser
Subject: A start job for unit serial-getty@ttyS0.service has finished successful
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

A start job for unit serial-getty@ttyS0.service has finished successfully.

The job identifier is 115066.
```

3.24 Запуск DNS-сервера

Служба named перезапущена и перешла в состояние active.

```
root@server:~# systemctl restart named
root@server:~# systemctl is-active named
active
root@server:~#
```

4. Результаты

4.1 Проверка прямой зоны

Команда `dig ns.rapavlichenko.net` вернула авторитетный ответ 192.168.1.1.

```
root@server:~# dig ns.rapavlichenko.net

; <<>> DiG 9.18.33 <<>> ns.rapavlichenko.net
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 47754
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 569564e0ce870429010000006aa9b550c4bb764e4f3084f2 (good)
;; QUESTION SECTION:
;ns.rapavlichenko.net.          IN      A

;; ANSWER SECTION:
ns.rapavlichenko.net.  86400   IN      A      192.168.1.1

;; Query time: 0 msec
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
;; WHEN: Tue Sep 15 21:14:56 UTC 2026
;; MSG SIZE  rcvd: 93
```

4.2 Проверка записей зоны

Команды host подтвердили работу A-, NS-, SOA- и PTR-записей.

```
root@server:~# host -l rapavlichenko.net
rapavlichenko.net name server rapavlichenko.net.
rapavlichenko.net has address 192.168.1.1
ns.rapavlichenko.net has address 192.168.1.1
server.rapavlichenko.net has address 192.168.1.1
root@server:~# host -a rapavlichenko.net
Trying "rapavlichenko.net"
;; -->HEADER<<- opcode: QUERY, status: NOERROR, id: 20534
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;rapavlichenko.net.          IN      ANY

;; ANSWER SECTION:
rapavlichenko.net.  86400  IN      SOA      rapavlichenko.net. server.rapavlichenko.net. 2026091500 86400 3600 604800 108
rapavlichenko.net.  86400  IN      NS       rapavlichenko.net.
rapavlichenko.net.  86400  IN      A        192.168.1.1

Received 108 bytes from 127.0.0.1#53 in 0 ms
root@server:~# host -t A rapavlichenko.net
rapavlichenko.net has address 192.168.1.1
root@server:~# host -t PTR 192.168.1.1
1.1.168.192.in-addr.arpa domain name pointer server.rapavlichenko.net.
1.1.168.192.in-addr.arpa domain name pointer ns.rapavlichenko.net.
root@server:~#
```

5. Автоматизация

5.1 Сохранение рабочей конфигурации

Файлы BIND и зоны скопированы в каталог `provision/server/dns` для последующего автоматического развёртывания.

```
vagrant@server:~$ sudo -i
root@server:~# cd /vagrant
root@server:/vagrant# mkdir -p /vagrant/provision/server/dns/etc/named
root@server:/vagrant# mkdir -p /vagrant/provision/server/dns/var/named/master
root@server:/vagrant# cp -R /etc/named.conf /vagrant/provision/server/dns/etc/
root@server:/vagrant#
root@server:/vagrant# cp -R /etc/named/* /vagrant/provision/server/dns/etc/named/
root@server:/vagrant# cp -R /var/named/master/* /vagrant/provision/server/dns/var/named/master/
root@server:/vagrant#
```

5.2 Сценарий настройки DNS

Сценарий `dns.sh` устанавливает пакеты, размещает конфигурацию, настраивает права, SELinux, `firewalld` и `NetworkManager`, затем запускает `named`.

```
GNU nano 8.1 /vagrant/provision/server/dns.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install bind bind-utils
echo "Copy configuration files"
cp -R /vagrant/provision/server/dns/etc/* /etc
cp -R /vagrant/provision/server/dns/var/named/* /var/named
chown -R named:named /etc/named
chown -R named:named /var/named
restorecon -vR /etc
restorecon -vR /var/named
echo "Configure firewall"
firewall-cmd --add-service=dns
firewall-cmd --add-service=dns --permanent
echo "Tuning SELinux"
setsebool named_write_master_zones 1
setsebool -P named_write_master_zones 1
echo "Change dns server address"
nmcli connection edit "System eth0" <<EOF
remove ipv4.dns
set ipv4.ignore-auto-dns yes
set ipv4.dns 127.0.0.1
save
quit
EOF
systemctl restart NetworkManager
echo "Start named service"
systemctl enable named
systemctl start named
```

5.3 Подключение сценария в Vagrantfile

Provisioner server dns запускает сценарий при развёртывании серверной виртуальной машины.

```
server.vm.network :private_network,  
  ip: "192.168.1.1",  
  virtualbox____intnet: true  
  
server.vm.provision "server dummy",  
  type: "shell",  
  preserve_order: true,  
  path: "provision/server/01-dummy.sh"  
  
server.vm.provision "server forwarding",  
  type: "shell",  
  preserve_order: true,  
  path: "provision/server/02-forward.sh"  
  
server.vm.provision "server dns",  
  type: "shell",  
  preserve_order: true,  
  path: "provision/server/dns.sh"  
  
end  
  
config.vm.define "client", autostart: false do |client|  
  client.vm.box = "rockylinux10"  
  client.vm.hostname = "client"  
  client.vm.boot_timeout = 1440
```

6. Выводы

6.1 Выводы

В ходе работы были получены практические навыки установки и настройки DNS-сервера BIND. Были созданы прямая и обратная зоны, настроены сетевые параметры, firewalld и SELinux, проверено разрешение имён и подготовлена автоматизация конфигурации сервера средствами Vagrant.

7. Завершение

Спасибо за внимание